A1 Problems: Numbers and divisibility

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No calculator, unless a problem says otherwise. For the proofs, write full sentences: a reader should be able to follow every step.

Practice

1. Calculate.

(a) $-7 + 12 - (-5)$
(b) $(-3) \cdot (-4) \cdot (-2)$
(c) $24 \div (-6) + (-2)^2$
(d) -2^2 and $(-2)^2$. Explain why they are different.
(e) $(-1)^{2027} + (-1)^{2026}$

2. Calculate, showing the order of operations (*rekenvolgorde*) you use.

(a) $3 + 4 \cdot 5 - 6 \div 2$
(b) $2 \cdot (3 + 4)^2 - 10 \div 5$
(c) $48 \div 4 \div 2 \cdot 3$
(d) $-3^2 + 4 \cdot (-1)^3 - (5 - 8)$

3. Without dividing, decide whether 7356 is divisible by 2, 3, 4, 5, 6, 8, 9 and 11. Give the rule you used each time.

4. Write as a product of primes (*ontbind in priemfactoren*), using powers where possible.

(a) 360 (b) 1001 (c) 2310 (d) 4096 (e) 2024

5. Use prime factorisations to find $\gcd(84, 120)$ and $\operatorname{lcm}(84, 120)$. Check that $\gcd \cdot \operatorname{lcm} = 84 \cdot 120$.

6. Use Euclid's algorithm to find

(a) $\gcd(2024, 1771)$ (b) $\gcd(252, 198)$.

7. Use the sieve of Eratosthenes to list all primes below 100. How many are there? Explain why you only needed to cross out multiples of 2, 3, 5 and 7.

8. (a) Write 72 as a product of primes. (b) Use it to say how many divisors 72 has, **before** listing them. (c) List them all.

9. Find the smallest positive whole number that is divisible by every number from 1 to 10.

School-exam level

10. Two buses leave the station at 08:00. Line 3 leaves every 12 minutes, line 7 every 18 minutes. When do they next leave at the same time? How many times do they leave together between 08:00 and 12:00 (both included)?

11. A floor of 84 cm by 60 cm is to be covered with equal square tiles, without cutting any tile. What is the largest possible tile, and how many such tiles are needed?

12. (a) The number $4\square52$ is divisible by 9. Which digit is $\square$?

(b) The five-digit number $2a63b$ (digits a and b) is divisible by 72. Find a and b.

13. (a) Show that the sum of any three consecutive integers is divisible by 3. (Hint: call the middle one n.)

(b) Show that the product of two consecutive integers is always even.
(c) Is the sum of four consecutive integers ever divisible by 4? Explain.

14. Prove the rule for 4: a whole number is divisible by 4 exactly when the number formed by its last two digits is divisible by 4. (Use $100 = 4 \cdot 25$.)

15. At 06:00 the temperature is $-8\,^\circ$C. It rises 3 degrees per hour until 13:00, then falls 2 degrees per hour.

(a) What is the temperature at 13:00 and at 22:00?
(b) At what two times is the temperature exactly $0\,^\circ$C?

16. (a) Find $\gcd(36, 60)$ and $\operatorname{lcm}(36, 60)$, and check that their product is $36 \cdot 60$.

(b) Explain, using prime factorisations, why $\gcd(a, b) \cdot \operatorname{lcm}(a, b) = ab$ always.
(c) Two numbers have gcd 6 and lcm 180. One of them is 36. What is the other?

17. A number leaves remainder 1 when divided by 2, by 3, by 4, by 5 and by 6, and it is divisible by 7. What is the smallest such number?

Challenge

18. Prove that $n^3 - n$ is divisible by 6 for every integer n. (Hint: factorise $n^3 - n = (n-1)\,n\,(n+1)$.)

19. Prove the rule for 11: a number is divisible by 11 exactly when its alternating digit sum (from the right: $+, -, +, -, \ldots$) is divisible by 11. Do it for six-digit numbers; explain why the same proof works for any length. (Hint: $10 = 11 - 1$, $100 = 99 + 1$, $1000 = 1001 - 1$, ...)

20. (a) Show that every odd number is of the form $4k + 1$ or $4k + 3$, and that the product of two numbers of the form $4k + 1$ is again of the form $4k + 1$.

(b) Prove that there are infinitely many primes of the form $4k + 3$. (Hint: given a list of such primes $p_1, \ldots, p_m$, look at $N = 4p_1p_2 \cdots p_m - 1$.)

21. Prove that if p is a prime greater than 3, then $p^2 - 1$ is divisible by 24.

22. How many zeros does $100! = 1 \cdot 2 \cdot 3 \cdots 100$ end in? Explain your reasoning.

23. Find all whole numbers $n \geq 0$ such that $n+1$ divides n^2+1. (Hint: $n^2+1 = (n+1)(n-1)+2$.)

24. The Fibonacci numbers are $1, 1, 2, 3, 5, 8, 13, \ldots$, each the sum of the two before.

(a) Use Euclid's algorithm to find $\gcd(377, 233)$. What do you notice about the quotients?
(b) Prove that any two consecutive Fibonacci numbers are coprime.

25. (a) Show that every perfect square leaves remainder 0 or 1 when divided by 4.

(b) Deduce that 2027 is not the sum of two perfect squares.
(c) Can a number ending in the digits ... 11 (such as 111, 2011, 34 511) be a perfect square?

26. A school has 100 lockers, numbered 1 to 100, all closed. Pupil 1 opens every locker. Pupil 2 changes (closes if open, opens if closed) every 2nd locker. Pupil 3 changes every 3rd locker, and so on, up to pupil 100. Which lockers are open at the end? Explain why.

27. (a) Use Euclid's algorithm to find $d = \gcd(1001, 221)$.

(b) Run the algorithm backwards to find integers x and y with $1001x + 221y = d$.
(c) Explain why there are no integers x, y with $1001x + 221y = 20$.